

Transformacije:

```
void CrobotView::Translate(CDC* pDC, float dx, float dy, bool rightMultiply)
{
    XFORM transform = { 1, 0, 0, 1, dx, dy };

    pDC->ModifyWorldTransform(&transform, rightMultiply ? MWT_RIGHTMULTIPLY :
    MWT_LEFTMULTIPLY);
}

void CrobotView::Scale(CDC* pDC, float sx, float sy, bool rightMultiply)
{
    XFORM transform = { sx, 0, 0, sy, 0, 0 };

    pDC->ModifyWorldTransform(&transform, rightMultiply ? MWT_RIGHTMULTIPLY :
    MWT_LEFTMULTIPLY);
}

void CrobotView::Rotate(CDC* pDC, float angle, bool rightMultiply)
{
    XFORM transform = { cos(angle), sin(angle), -sin(angle), cos(angle) };

    pDC->ModifyWorldTransform(&transform, rightMultiply ? MWT_RIGHTMULTIPLY :
    MWT_LEFTMULTIPLY);
}

void CrobotView::Mirror(CDC* pDC, bool mx, bool my, bool rightMultiply)
{
    Scale(pDC, mx ? -1 : 1, my ? -1 : 1, rightMultiply);
}
```

Fliker:

```
CRect rect;

GetClientRect(&rect);
CDC* memDC = new CDC();

memDC->CreateCompatibleDC(pDC);

CBitmap bmp;

bmp.CreateCompatibleBitmap(pDC, rect.Width(), rect.Height());

CBitmap* oldbmp = memDC->SelectObject(&bmp);
```

```
memDC->SetGraphicsMode(GM_ADVANCED);
XFORM transOld;

memDC->GetWorldTransform(&transOld);
//Funkcije za crtanje

DrawRobot(memDC);

pDC->BitBlt(0, 0, rect.Width(), rect.Height(), memDC, 0, 0, SRCCOPY);

memDC->SelectObject(oldbmp);

memDC->SetWorldTransform(&transOld);
delete memDC;
```

Klikovi na dugme:

```
void CrobotView::OnKeyDown(UINT nChar, UINT nRepCnt, UINT nFlags)
{
// TODO: Add your message handler code here and/or call default
if (nChar == 'A')
{
if (sakaRot > -10 * toRad)

sakaRot -= 5 * toRad;
}
else if (nChar == 'S')
{
if (sakaRot < 30 * toRad)

sakaRot += 5 * toRad;
}
else if (nChar == 'D')
{
if (podlakticaRot > -10 * toRad)

podlakticaRot -= 5 * toRad;
}
else if (nChar == 'F')
{
if (podlakticaRot < 80 * toRad)

podlakticaRot += 5 * toRad;
}
}
```

```

else if (nChar == 'G')
{
if (nadlakticaRot > -10 * toRad)

nadlakticaRot -= 5 * toRad;
}
else if (nChar == 'H')
{
if (nadlakticaRot < 90 * toRad)

nadlakticaRot += 5 * toRad;
}
Invalidate();

CView::OnKeyDown(nChar, nRepCnt, nFlags);
}

BOOL CrobotView::OnEraseBkgnd(CDC* pDC)
{
// TODO: Add your message handler code here and/or call default
return TRUE; // ovo mora da bi fliker radio kako treba
return CView::OnEraseBkgnd(pDC);
}

```

Ovo mora da bi radile transformacije:

```

int prevMode = SetGraphicsMode(pDC->m_hDC, GM_ADVANCED);
XFORM xFormOld;

pDC->GetWorldTransform(&xFormOld);

pDC->SetWorldTransform(&xFormOld);

pDC->SetGraphicsMode(prevMode);

```

Ucitavanje aleta:

Slike:

```

DImage* naziv; // ovo ide u .h
naziv = new DImage(); // konstruktor u .cpp
naziv->Load(CString("naziv.png")); // isto konstruktor u .cpp
delete glava; // destruktor u .cpp

```

Meta fajlovi:

```
HENHMETAFILE naziv; // .h fajl
```

```
naziv = GetEnhMetaFile(CString("naziv.emf")); // .cpp konstruktor
```

```
this->naziv = GetEnhMetaFile(CString("naziv.emf"));
```

```
DeleteEnhMetaFile(naziv); // .cpp destruktorktor
```

```
//f-ja
```

```
ENHMETAHEADER header;
```

```
GetEnhMetaFileHeader(mf_viljuska, sizeof(ENHMETAHEADER), &header);
```

```
int w = header.rc1Bounds.right - header.rc1Bounds.left;
```

```
int h = header.rc1Bounds.bottom - header.rc1Bounds.top;
```

```
...
```

```
PlayEnhMetaFile(pDC->m_hDC, mf_viljuska, CRect(0, 0, w, h));
```

```
//Ili samo:
```

```
pDC->PlayMetaFile(this->naziv, CRect(-15, 0, 15, -37.5));
```

```
//Olovke
```

```
CPen* kaktusOlovka = new CPen(PS_SOLID, 1, RGB(0, 0, 0));
```

```
// CETKE
```

```
CBrush* pozadinskaCetka = new CBrush(RGB(135, 206, 235));
```